Rule Based Heat Exchanger Monitoring Implementation and Synthetic Scenario Evaluation

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Abstract*—Heat exchanger measurements can indicate reduced thermal performance without establishing its cause. This study documents a Python implementation that combines basic physical checks, heat balance calculations, and ordered screening rules. The program reads operating records, calculates heat duties, terminal temperature differences, overall heat transfer coefficients, and pressure drops, and evaluates ratios relative to a designated healthy observation. Four manually constructed records exercise normal operation, a fouling-like pattern, reduced hot-side flow, and an invalid temperature measurement. The three valid records yield overall heat transfer coefficients of 267.06, 190.62, and 173.91 watts per square metre kelvin. The fourth record fails the terminal temperature check and is retained with an invalid-data label. The repository contains 17 automated tests, which were rerun for this revision. Agreement with the intended scenario labels establishes that the implementation executes the specified rules. It does not measure fault detection accuracy because the scenarios were constructed for those rules and contain no independent fault observations. The analysis identifies limitations of fixed reference ratios, incomplete input validation, rule precedence, and the treatment of energy imbalance. Further evaluation requires independent operating records, operating-condition adjustment, measurement uncertainty, and maintenance or inspection evidence.*

Index Terms*—condition monitoring, heat exchangers, rule-based systems, software verification, synthetic data.*

I. INTRODUCTION

Heat exchanger monitoring requires a distinction between observed performance and the cause of a change. The heat balance and log-mean temperature difference provide useful performance indicators [1], but fouling assessment requires attention to operating conditions and hydraulic behavior. Diaz-Bejarano et al. [2] combined thermal and pressure measurements with dynamic models to analyze refinery fouling and cleaning. Their work shows why thermal measurements alone can leave deposit properties poorly constrained.

Subsequent work by the same authors [3] examined limitations of conventional indicators and developed a thermal–hydraulic visualization method for variable operating conditions. Their analysis extends beyond a comparison with a single clean observation: it uses exchanger models to interpret changes in performance. Ujević Andrijić and Rimac [4] instead used neural networks and gradient boosting to estimate clean outlet temperatures from refinery operating data. Their reported difficulty with extrapolation illustrates that a baseline model also needs evaluation outside its development conditions.

The present work addresses a narrower implementation question: can a small, inspectable program calculate the intended indicators, reject a physically inconsistent record, and assign the specified labels to constructed operating patterns? The contribution is an educational software implementation and a documented scenario evaluation. No new heat-transfer correlation, fouling mechanism, trained classifier, or industrial performance claim is proposed. The rules and software outputs are distinguished throughout from evidence of the underlying equipment condition.

II. METHODOLOGY

A. Input Data and Synthetic Scenarios

Each input row contains a timestamp, scenario label, hot- and cold-side mass flow rates, constant-pressure heat capacities, inlet and outlet temperatures, inlet and outlet pressures, and heat-transfer area. The demonstration data were manually constructed from simplified engineering relationships; they are neither copied from a plant nor generated by a trained model.

The four rows are test fixtures, not a sample from an operating population. They were selected to activate particular decision paths. Consequently, the scenario names describe design intent rather than independently observed fault labels. The timestamps are identifiers; the program does not fit trends or treat the rows as a degradation trajectory. All input values are provided in Appendix A.

TABLE I. MANUALLY CONSTRUCTED DEMONSTRATION SCENARIOS

Scenario	Designed pattern	Expected label
Healthy	Balanced duty; baseline U and ΔP	normal
Fouling	Lower U; higher ΔP	possible_fouling
Low flow	Lower hot flow and hot-side ΔP	possible_low_flow
Bad sensor	Impossible terminal temperature	invalid_data

B. Validation Layer

File-level validation requires all named columns. Row-level validation rejects missing values, duplicate timestamps, nonpositive flow rates, heat capacities or area, unexpected hot-side heating, unexpected cold-side cooling, nonpositive pressure drops, and nonpositive terminal temperature differences for counter-current flow. Invalid rows remain in the exported result for auditability, but their derived indicators are set to not-a-number (NaN).

The loader also parses timestamps and raises an error if parsing fails. These checks encode the assumed operating mode rather than every physically possible exchanger state. Isothermal phase change, flow reversal, startup, and negligible pressure loss require different acceptance criteria. The implementation has no complete numeric-type, finite-value, sensor-range, or unit-conversion validation layer; passing its checks therefore does not establish measurement reliability.

C. Thermal and Hydraulic Indicators

The hot- and cold-side heat duties are calculated as

$$Q_h = \dot{m}_h c_{p,h}(T_{h,in} - T_{h,out}), \quad Q_c = \dot{m}_c c_{p,c}(T_{c,out} - T_{c,in}) \tag{1}$$

where mass flow is in kg s^{-1} and heat capacity is in kJ kg^{-1} K^{-1}, yielding duty in kW. A symmetric energy-balance deviation is used:

$$\varepsilon_Q = \frac{|Q_h - Q_c|}{(Q_h + Q_c)/2} \times 100\% \tag{2}$$

For counter-current flow, define ΔT_1 = Th,in − Tc,out and ΔT_2 = Th,out − Tc,in. The log-mean temperature difference (LMTD) is

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)} \qquad (3)$$

The implementation uses the mean of the two measured duties to estimate UA and U:

$$\bar{Q} = \frac{Q_h + Q_c}{2}, \quad UA = \frac{\bar{Q}}{\Delta T_{lm}}, \quad U = \frac{1000\,UA}{A} \qquad (4)$$

The factor 1000 converts kW K^{-1} to W K^{-1}. Side-specific pressure drop is

$$\Delta P = P_{in} - P_{out} \qquad (5)$$

Here h and c denote the hot and cold streams, respectively; A is the stated heat-transfer area. The calculation assumes sensible heating and cooling with supplied constant heat capacities and a steady-state interpretation. It uses counter-current terminal differences without a multipass correction factor. If the terminal differences are approximately equal, the code uses their limiting value for LMTD to avoid division by a vanishing logarithm. Temperature differences in degrees Celsius and kelvin are numerically equal.

The arithmetic mean duty is an implementation choice, not statistical data reconciliation. A large discrepancy between the two duties is flagged later, but the program still exports the U calculated from their mean. Such a value should not be interpreted as a reliable equipment coefficient until the balance discrepancy has been investigated.

D. Reference Normalization and Screening Rules

The first valid row with scenario name healthy is used as the internal baseline. For each indicator x, its normalized ratio is

$$R_x = \frac{x}{x_{ref}} \times 100\% \qquad (6)$$

Rules are evaluated sequentially. Invalid data take precedence, followed by excessive energy imbalance, low-flow patterns, fouling-like patterns, reduced heat transfer, and normal operation. Consequently, each reported label has a deterministic explanation.

The thresholds of 10%, 80%, 85%, and 120% are hard-coded demonstration settings. They were not estimated from a measured distribution, optimized against labeled faults, or prescribed by an engineering standard. The inequalities are strict: equality at a threshold does not trigger that branch. The low-flow branch checks a flow ratio and its corresponding side-specific pressure-drop ratio; it does not require a reduced U. The scenario field selects the reference but does not directly supply the nonreference classification.

TABLE II. ORDERED RULE SET AND ILLUSTRATIVE THRESHOLDS

Rule	Demonstration criterion	Output
Validity	Any row-level check fails	invalid_data
Balance	εQ > 10%	energy_imbalance
Low flow	Flow < 80% and corresponding ΔP < 80%	possible_low_flow
Fouling-like	U < 85% and either ΔP > 120%	possible_fouling
Thermal loss	U < 85% without high-ΔP pattern	reduced_heat_transfer
Otherwise	No preceding criterion met	normal

III. SOFTWARE ARCHITECTURE AND VERIFICATION

The application is organized into modules for CSV loading, validation, calculations, analysis, diagnostics, pipeline orchestration, visualization, and command-line execution. Pandas supports tabular processing [5], while Matplotlib generates the monitoring figure [6]. The pipeline preserves a single, reproducible order: load, validate, calculate, normalize, screen, export, and visualize.



Fig. 1. Implemented monitoring pipeline.

The released repository contains 17 automated tests covering individual formulas, CSV loading, row validation, healthy-case analysis, invalid-row handling, diagnostic classifications, end-to-end pipeline execution, and figure generation. These tests establish repeatable software behavior against specified examples. They do not establish field sensitivity, specificity, robustness to sensor drift, or diagnostic accuracy under industrial operating variability.

The suite was rerun during preparation of this revision and all 17 tests passed. A passing suite is limited by its assertions: the supplied scenario test exercises four labels, whereas the energy_imbalance and reduced_heat_transfer branches are not demonstrated by the four-row dataset. Nor does the test count constitute a code-coverage percentage. Boundary cases, mixed faults, malformed numeric fields, infinite values, and changing operating regimes require additional evaluation.

IV. RESULTS AND DISCUSSION

Table III reports the exact outputs obtained from the version 1.0.0 implementation. All valid scenarios were deliberately constructed with equal hot- and cold-side duties; therefore, their energy-balance deviation is 0%. This idealized balance is a property of the synthetic examples and should not be expected from real measurements.

TABLE III. MODEL OUTPUTS (Q IN kW, LMTD IN °C, U IN W m^{-2} K^{-1}, ΔP IN kPa)

Case	Qh/Qc	LMTD	U (% ref.)	ΔPh/ΔPc	Status
Healthy	400/400	74.89	267.06 (100)	40/35	normal
Fouling	320/320	83.94	190.62 (71.38)	55/50	possible_fouling
Low flow	280/280	80.50	173.91 (65.12)	20/35	possible_low_flow
Bad sensor	—	—	—	—	invalid_data

Relative to the healthy reference, the fouling scenario has U/Uref = 71.38% (U = 190.62 W m^{-2} K^{-1}) and hot- and cold-side pressure-drop ratios of 137.5% and 142.86%. It therefore satisfies the low-U/high-pressure-drop rule. The low-flow scenario has a hot-flow ratio of 70% and a hot-side pressure-drop ratio of 50%, satisfying the paired low-flow/low-pressure-drop rule. The bad-sensor row has a cold outlet temperature above the hot inlet temperature, causing a nonpositive terminal temperature difference; it is labeled invalid_data before thermal calculations are attempted.

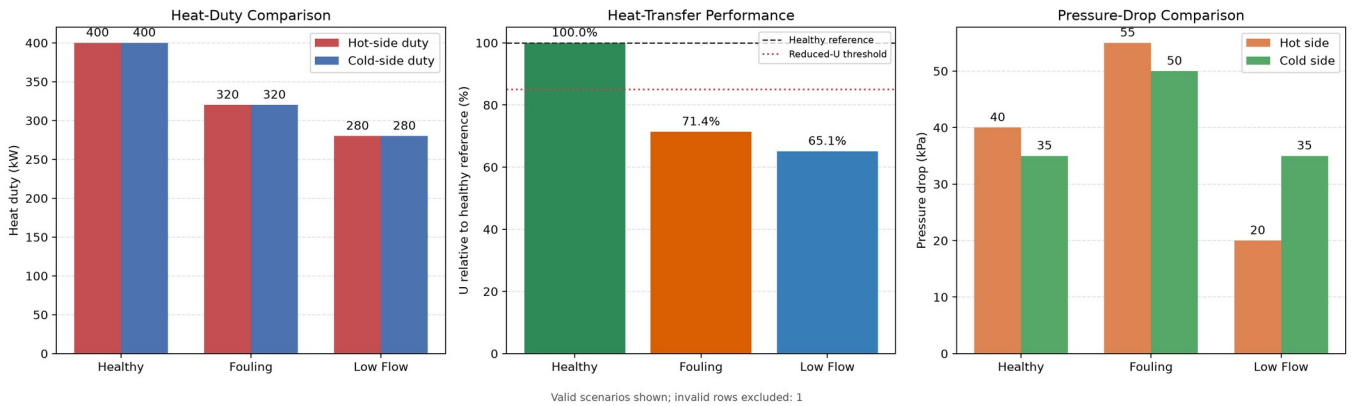


Fig. 2. Monitoring summary for the three valid synthetic scenarios; one invalid row is excluded from the charts.

Figure 2 visualizes the same deterministic outputs. The figure supports comparison but does not provide statistical evidence: the dataset contains one constructed row per scenario, no repeated measurements, and no uncertainty intervals.

A. Interpretation of the Constructed Cases

The fouling-like case reduces duty by 20% and apparent U by 28.62% while keeping both mass flow rates unchanged. The cold outlet temperature was selected to balance the specified hot-side duty. Its increased pressure differences were assigned independently; they were not calculated from a deposit-thickness or friction model. The example therefore demonstrates a pattern consistent with the implemented fouling rule, rather than a mechanistic simulation of deposition.

The low-flow case reduces hot-side mass flow from 2.0 to 1.4 kg s^{-1}. The hot temperature fall remains 50 K, so the hot duty falls from 400 to 280 kW. The cold outlet is set to 58 °C to match this duty. The assigned hot pressure drop falls from 40 to 20 kPa. These values test the paired low-flow criterion but do not establish a general quantitative relation between flow and pressure loss.

For the invalid record, the cold outlet is 155 °C and the hot inlet is 150 °C. Their terminal difference is −5 K, which violates the assumed counter-current sensible-heat model. The program detects a consistency failure; it cannot determine which sensor is faulty or whether another data problem caused the inconsistency. Conversely, a plausible but biased sensor reading may pass all current checks.

B. What the Evaluation Establishes

The observed label agreement follows from constructing the cases around the rule set. Reporting four correct classifications as an accuracy estimate would therefore be circular. There is no independent test population, blinded labeling, confidence interval, or evidence of performance on overlapping fault patterns. The current evidence supports formula reproduction, execution of selected decision paths, and production of the expected output artifacts.

V. LIMITATIONS AND INDUSTRIAL VALIDATION ROADMAP

The prototype has six principal limitations. First, four synthetic rows cannot characterize natural process variability. Second, heat capacities are supplied constants rather than temperature- and composition-dependent properties. Third, the framework does not model exchanger geometry, Reynolds, Prandtl, or Nusselt numbers. Fourth, it does not estimate fouling resistance or propagate measurement uncertainty. Fifth, the thresholds are illustrative settings applied against one synthetic reference. Sixth, no labels have been checked against inspection, cleaning, or maintenance outcomes.

A defensible industrial study would require: 1) governed time-series historian data covering multiple operating regimes; 2) verified units, sampling alignment, sensor calibration, exchanger geometry, and fluid properties; 3) regime-specific healthy baselines; 4) ground truth from work orders, inspections, cleaning events, and engineering review; 5) uncertainty and sensitivity analysis; and 6) prospective evaluation of false alarms, missed events, lead time, and operational value. Only after these steps should the rules be calibrated or compared with statistical and machine-learning alternatives.

A. Reference Dependence and Diagnostic Ambiguity

The main modeling limitation is the fixed baseline. A clean exchanger can exhibit a different apparent U when flow rates or fluid properties change. Pressure loss also responds to operating conditions. Ratios against one record do not remove these effects. The literature in [2] and [3] motivates adjustment for operating conditions; the current implementation does not perform that adjustment. Its U ratio is therefore a relative indicator, not an estimate of fouling resistance or thermal effectiveness.

Rule precedence also limits interpretation. If an observation satisfies both the low-flow and fouling criteria, the low-flow label is returned because that branch is evaluated first. No probabilities or concurrent labels are reported. Similarly, normal means only that no earlier rule was activated; it is not a declaration that the equipment is fault-free. Flow restriction, valve changes, bypassing, and measurement bias can produce ambiguous combinations of indicators.

B. Proposed Evaluation Protocol

A useful next study would first define the equipment and operating envelope, then assemble stable operating windows with synchronized measurements. Clean periods should be supported by inspection or cleaning records rather than by a scenario name alone. A baseline should be estimated from development periods and evaluated on later, separate periods, including changes in throughput and inlet temperature. Fault labels require independent maintenance or experimental evidence. Thresholds must be selected before assessment on the held-out periods.

Assessment should report event-level missed detections, false alarms per operating time, detection delay, and results by operating regime. Measurement uncertainty should be propagated through duty and LMTD calculations, particularly where a small terminal temperature difference amplifies error. Threshold sensitivity and mixed-fault cases should be examined. These are

proposed evaluation steps; none has been completed for the present prototype.

Machine learning would require a separate development study. The clean-temperature prediction approach in [4] offers one possible comparison, but its findings cannot be transferred to this dataset. Any comparison should use the same time-separated observations, operating envelope, and independent reference labels. Four constructed records provide no basis for training or comparing predictive models.

VI. CONCLUSION

An explainable heat-exchanger monitoring MVP was implemented as a modular Python pipeline. The software validates measurements, calculates thermal and hydraulic indicators, compares valid operation with a healthy reference, applies ordered diagnostic rules, and exports auditable results and a summary figure. The constructed scenarios activate the intended normal, possible-fouling, possible-low-flow, and invalid-data paths, and 17 automated tests verify the programmed behavior. The study therefore demonstrates implementation feasibility and traceability, not industrial diagnostic validity. The next scientific milestone is validation against real or experimental time-series data with maintenance-based ground truth and quantified uncertainty.

REPRODUCIBILITY STATEMENT

Source code, input data, tests, and example outputs are available in [7]. The implementation examined here is the v1.0.0 release associated with commit 165813f. In an environment with the declared dependencies installed, execute the test and analysis commands given below. The default outputs are outputs/analysis_results.csv and outputs/monitoring_summary.png. Dependency versions in the repository are not a complete environment lock; exact long-term reproduction also requires recording the Python and package versions.

```
PYTHONPATH=src pytest -q
PYTHONPATH=src python -m \
  heat_exchanger_monitoring.cli \
  data/synthetic_operating_data.csv
```

Backslashes continue the analysis command onto the next line.

APPENDIX A COMPLETE SCENARIO INPUTS

All records use a heat-transfer area of 20 m² and hot- and cold-side heat capacities of 4.0 kJ kg⁻¹ K⁻¹. Inlet pressures are 500 kPa on the hot side and 400 kPa on the cold side. The scenario timestamps on 1 November 2026 are 08:00, 09:00, 10:00, and 11:00, respectively. The remaining values are listed below. They reproduce the supplied CSV without implying a particular fluid, exchanger geometry, or measured plant condition.

TABLE IV. COMPLETE VARIABLE INPUTS FOR THE FOUR RECORDS

Input	Healthy	Fouling	Low flow	Bad sensor
Hot flow (kg/s)	2.0	2.0	1.4	2.0
Cold flow (kg/s)	2.5	2.5	2.5	2.5
Hot inlet (°C)	150	150	150	150
Hot outlet (°C)	100	110	100	100
Cold inlet (°C)	30	30	30	30
Cold outlet (°C)	70	62	58	155
Hot outlet (kPa)	460	445	480	460
Cold outlet (kPa)	365	350	365	365

For example, the healthy hot duty is $2.0 \times 4.0 \times (150 - 100) = 400$ kW; the cold duty is $2.5 \times 4.0 \times (70 - 30) = 400$ kW. Terminal differences of 80 and 70 K give an LMTD of 74.8888 K. Dividing 400 kW by this value and by 20 m², with the unit conversion in (4), gives $U = 267.0628$ W m^{-2} K^{-1}. This independent arithmetic check provides a transparent reference for the first valid record.

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REFERENCES

[1] R. K. Shah and D. P. Sekulić, Fundamentals of Heat Exchanger Design. Hoboken, NJ, USA: Wiley, 2003.

[2] E. Diaz-Bejarano, F. Coletti, and S. Macchietto, “Thermo-hydraulic analysis of refinery heat exchangers undergoing fouling,” AIChE J., vol. 63, no. 3, pp. 984–1001, 2017, doi: 10.1002/aic.15457.

[3] E. Diaz-Bejarano, F. Coletti, and S. Macchietto, “A model-based method for visualization, monitoring, and diagnosis of fouling in heat exchangers,” Ind. Eng. Chem. Res., vol. 59, no. 10, pp. 4602–4619, 2020, doi: 10.1021/acs.iecr.9b05490.

[4] Ž. Ujević Andrijić and N. Rimac, “Data-driven fouling detection in refinery preheat train heat exchangers using neural networks and gradient boosting,” Sensors, vol. 25, no. 16, Art. no. 4936, 2025, doi: 10.3390/s25164936.

[5] W. McKinney, “Data structures for statistical computing in Python,” in Proc. 9th Python Sci. Conf., Austin, TX, USA, 2010, pp. 56–61.

[6] J. D. Hunter, “Matplotlib: A 2D graphics environment,” Comput. Sci. Eng., vol. 9, no. 3, pp. 90–95, May–Jun. 2007, doi: 10.1109/MCSE.2007.55.

[7] G. Huseynli, “Heat Exchanger Performance Monitoring and Rule-Based Fault Screening,” GitHub repository, release v1.0.0, Nov. 2026. [Online]. Available: https://github.com/gashamhuseynli/heat-exchanger-monitoring